



Modelling Financial Markets

Seminar 1: Stylised Facts and Returns — Hands-on



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Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CI: Confidence Interval
- COVID-19: COronaVirus Disease 2019
- CUSUM: CUmulative SUM (of squares)
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- FIGARCH: Fractionally Integrated GARCH
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- JB: Jarque–Bera (test)
- MDD: Maximum DrawDown
- OHLC: Open, High, Low, Close
- OLS: Ordinary Least Squares
- QQ: Quantile–Quantile (plot)
- RMSE: Root Mean Squared Error
- SE: Standard Error

Acronyms used in this material (2/2)

- **US: United States**
- **UTC: Coordinated Universal Time (Temps Universel Coordonné)**
- **VIX: CBOE Volatility Index**

Today's Question and Route (1/2)

- Question of the seminar
 - ▶ **which stylised facts of returns survive inference that allows for heavy tails and volatility clustering?**
- Route
 - ▶ prerequisites for today: six primer slides (the lecture comes after this seminar)
 - ▶ returns, drawdowns and moments: A1, A4(a), B1, A3, B2
 - ▶ tails and dependence: A6, B3, B9, B4
 - ▶ an AI answer to correct: C2
 - ▶ after the lecture: A7, B11, A10

Today's Question and Route (2/2)

- Outputs to produce
 - ▶ the A answers
 - ▶ one figure and one table per B exercise
 - ▶ the corrected C2 code
 - ▶ one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#)
 - ▶ each task names its notebook section
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how do returns add up over time?	Solved	–
A4(a)	how deep is a drawdown?	Proposed	A1
B1	how heavy are the tails, and how fragile is the kurtosis?	Solved	–
A3	from a statistic to a p-value: JB and Ljung–Box	Solved	–
B2	how large is the VaR 1% (Value-at-Risk) of BET-TR?	Proposed	B1
A6	why does random volatility create heavy tails?	Solved	–
B3	are Bitcoin returns independent?	Solved	–
B9	which autocorrelations survive robust inference?	Solved	–
B4	do returns approach the Normal distribution with the horizon?	Proposed	B1, B3
C2	what is wrong in an AI answer?	Proposed	A3, B3, B9
After the lecture	A7 [Solved], B11 [Solved], A10 [Proposed] (tail heaviness, width of the ACF band)	homework	A6, B9
Extension packs	A1(c), A2, A4(b–e), A5, A8, A9, A11, B1(d), B5–B8, B10, B12–B15, C1, C3 (optional)	Solved or Proposed (marked on each)	named on each task

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy
- ▣ **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper
- ▣ Part B: verification and inference on real data, each ending with an interpretation question
- ▣ Part C: open questions and AI critique

Data Used

Series	Price used	Period	Calendar	Returns
BET-TR	daily close	23 Sep 2014 – 18 Sep 2026	BVB trading days	2 992
Bitcoin	daily close (USD)	17 Sep 2014 – 18 Sep 2026	every day, 24:00 UTC	4 384
EUR/RON	BNR reference rate	4 Jul 2005 – 18 Sep 2026	Romanian working days	5 352
S&P 500	daily OHLC	2 Jan 1990 – 18 Sep 2026	US trading days	9 245
SPY ETF	OHLC, adjusted close (for dividends and splits)	2 Jan 2008 – 18 Sep 2026 (B8)	US trading days	–
VIX, 9-day VIX	daily close	C3 only	US trading days	–

- Daily prices of the course data, one series per asset (field used: close / adjusted close)
- EUR/RON is the official BNR reference rate, not a series of market quotes
- Units: returns in decimals in the code, reported in % on the slides
- Calendars: each series on its own calendar
 - ▶ multi-series analyses first align the *prices* on common days

Setup: Run and Verify

```
# notebook section "Data" defines load_data(name)
closes = {a: load_data(a)['Close'] for a in
           ['sp500', 'bettr', 'btc', 'eurron']}
rets = {a: np.log(c).diff().dropna()
        for a, c in closes.items()}
r_bet, r_btc = rets['bettr'], rets['btc']
r_eur, r_sp = rets['eurron'], rets['sp500']
for a, r in rets.items():
    print(a, r.index[0].date(), len(r))
```

- Expected output: first return date and count

Series	first return	T
S&P 500	1990-01-03	9 245
BET-TR	2014-09-24	2 992
Bitcoin	2014-09-18	4 384
EUR/RON	2005-07-05	5 352

- Conventions in the code
 - moments: `scipy.stats.skew` and `kurtosis` (excess kurtosis, no small-sample correction)
 - standard deviations with `ddof=1`
 - quantiles with `numpy.quantile` (linear)
- If a count differs, stop and check the series, the price type and the start date

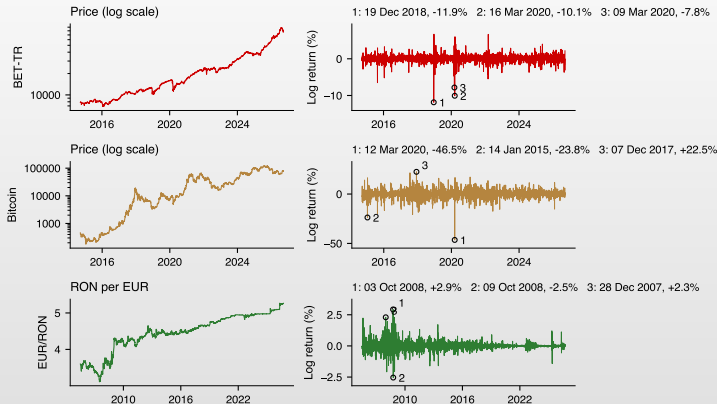
Notation Used in Every Exercise (1/2)

- P_t : price; T : number of returns
 - ▶ simple return $R_t = P_t/P_{t-1} - 1$
 - ▶ log return $r_t = \ln(P_t/P_{t-1})$
 - ▶ one worked line: $P_0 = 100$, $P_1 = 102$: $R_1 = 2.00\%$, $r_1 = \ln 1.02 = 1.98\%$
- $\bar{r}$, $\hat{\sigma}$: sample mean and standard deviation
 - ▶ standardised return $z_t = (r_t - \bar{r})/\hat{\sigma}$
 - ▶ demeaned return $x_t = r_t - \bar{r}$
- **i.i.d.**: independent and identically distributed
 - ▶ the reference assumption of the classical tests

Notation Used in Every Exercise (2/2)

- ▣ Moments of the standardised returns
 - ▶ **skewness** $\hat{S} = \frac{1}{T} \sum_t z_t^3$
 - ▶ **kurtosis** $\hat{K} = \frac{1}{T} \sum_t z_t^4$
 - ▶ **excess kurtosis** $\hat{K} - 3$ (0 for the Normal distribution)
- ▣ **Autocorrelation** at lag k : $\hat{\rho}_k = \sum_{t=k+1}^T x_t x_{t-k} / \sum_{t=1}^T x_t^2$
- ▣ **Annualisation** with P return intervals a year
 - ▶ $P = 252$ (BET-TR, S&P 500, EUR/RON), $P = 365$ (Bitcoin)
 - ▶ annual volatility $\sqrt{P} \hat{\sigma}$
- ▣ **p-value**: probability, under H_0 , of a statistic at least as extreme as the one observed
 - ▶ it is not the probability that H_0 is true

First Look: Prices, Returns and Dated Extremes



- Every extreme day is a dated event
 - ▶ the bank-asset tax (19 Dec 2018)
 - ▶ the COVID-19 crash (March 2020)
 - ▶ the 2008 depreciation of the leu
- Calm and turbulent periods alternate in all three series
- EUR/RON is almost flat after 2013
- Look first, test later

Prerequisites for Today (1/6): The Stylised Facts

- ▣ This seminar comes **before** the lecture
 - ▶ these six slides give only what the in-class tasks use
 - ▶ depth in [Chapter 1](#), the lecture after this seminar
- ▣ **Stylised fact:** a statistical property shared by many assets, markets and periods [Cont \(2001\)](#)
- ▣ Today's four stylised facts
 - ▶ **heavy tails:** extreme days are more frequent than under the Normal distribution with the same variance
 - ▶ **volatility clustering:** large moves follow large moves, of either sign
 - ▶ **no linear autocorrelation:** today's return is not a linear function of past returns, $\hat{\rho}_k \approx 0$
 - ▶ **aggregational Gaussianity:** returns over longer horizons have tails closer to those of the Normal distribution

Prerequisites for Today (2/6): Kurtosis and the Jarque–Bera Test

- **Heavy tails:** measured by the excess kurtosis $\hat{K} - 3$, 0 for the Normal distribution
 - ▶ one extreme day can move $\hat{K}$ a lot
 - ▶ robust alternative: the Crow–Siddiqui quantile excess kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$ Kim & White (2004)
 - ▶ it is also 0 for the Normal distribution
- **Jarque–Bera** Jarque & Bera (1980): $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$
 - ▶ H_0 : i.i.d. returns with the Normal distribution
 - ▶ under H_0 , $JB \sim \chi^2_2$; 5% critical value 5.99
- Example: $T = 1\,000$, $\hat{S} = 0$, $\hat{K} = 6$: $JB = 1\,000 \times 9/24 = 375 > 5.99$, reject H_0

Prerequisites for Today (3/6): Value-at-Risk and the Student- t

- ▣ **VaR 1%:** the one-day loss exceeded on 1% of days
 - ▶ $\text{VaR}_\alpha = -q_\alpha$, with q_α the α -quantile of the return
- ▣ Normal distribution: $q_\alpha = \mu + \sigma \Phi^{-1}(\alpha)$, $\Phi^{-1}(0.01) = -2.326$
 - ▶ example: $\mu = 0.05\%$, $\sigma = 1\%$: $q_{0.01} = 0.05 - 2.326 = -2.28\%$, so $\text{VaR } 1\% = 2.28\%$
- ▣ **Student- t** with ν degrees of freedom: $P(|X| > x)$ decays like $x^{-\nu}$
 - ▶ $\mathbb{E}|X|^p$ is finite only for $p < \nu$ (variance: $\nu > 2$; kurtosis: $\nu > 4$)
 - ▶ $t_3^{-1}(0.01) = -4.54$; rescaled to unit variance, $-4.54\sqrt{1/3} = -2.62$, deeper than -2.33
 - ▶ fitted by maximum likelihood with a location m and a scale s : `scipy.stats.t.fit`
- ▣ **Empirical VaR:** minus the sample 1% quantile
 - ▶ on the sample that produced it, it is breached on 1% of days by construction

Prerequisites for Today (4/6): ACF, Ljung–Box and ARCH-LM

- ▣ **ACF**: the autocorrelations $\hat{\rho}_k$, $k = 1, 2, \dots$
 - ▶ for i.i.d. returns $\hat{\rho}_k \approx N(0, 1/T)$
 - ▶ hence the band $\pm 1.96/\sqrt{T}$ ($T = 2500$: ± 0.039)
- ▣ **Volatility clustering** in the ACF: $|r_t|$ and r_t^2 are autocorrelated although r_t is not
- ▣ **Ljung–Box** [Ljung & Box \(1978\)](#): $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$, compared with χ_m^2 under i.i.d. returns
 - ▶ on r_t it tests linear predictability
 - ▶ on r_t^2 it tests volatility clustering
- ▣ **ARCH-LM** [Engle \(1982\)](#): regress x_t^2 on a constant and $x_{t-1}^2, \dots, x_{t-q}^2$
 - ▶ $LM = (T-q)R^2$, compared with χ_q^2
 - ▶ example: $T = 1000$, $q = 5$, $R^2 = 0.02$: $LM = 995 \times 0.02 = 19.9 > 11.07$, the 5% critical value of χ_5^2 : clustering

Prerequisites for Today (5/6): The Effect of Clustering on Inference

- ▣ **Mixture of Normal distributions** Clark (1973): $r_t = \sigma_t \varepsilon_t$ with random σ_t
 - ▶ kurtosis $3 \mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2 \geq 3$: heavy tails from Normal shocks (A6)
- ▣ Under clustering $\text{Var}(\hat{\rho}_k) > 1/T$
 - ▶ estimate it by $\hat{\tau}_k = \sum_{t>k} x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ Lobato, Nankervis & Savin (2001)
 - ▶ robust band $\pm 1.96 \sqrt{\hat{\tau}_k}$
 - ▶ robust $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$, compared with χ_m^2 (B9)
 - ▶ example: $T \hat{\tau}_1 = 4$ makes the robust band $\sqrt{4} = 2$ times wider than $\pm 1.96 / \sqrt{T}$

Prerequisites for Today (6/6): Aggregation and the QQ Plot

- ▣ **Aggregational Gaussianity:** an h -day log return is the sum of h daily ones
 - ▶ for i.i.d. returns with daily excess kurtosis κ , the h -day excess kurtosis is $\kappa/h \rightarrow 0$
- ▣ **QQ plot:** sorted standardised returns against the quantiles of the Normal distribution
 - ▶ a straight line means the Normal distribution
 - ▶ an S shape means heavy tails
- ▣ After the lecture
 - ▶ A7 and B11 estimate the tail exponent α in $P(|r_t| > x) \propto x^{-\alpha}$
 - ▶ A10 derives the variance of $\hat{\rho}_k$

A1 [Solved]: Returns and Annualisation — Task

▣ Questions

- ▶ which return adds up over time?
- ▶ how do daily numbers become annual ones?

▣ Inputs

- ▶ a share closes at 100, 110, 99, 104.5 on days 0–3 and pays no dividend
- ▶ an index has a daily mean log return of 0.04% and a daily volatility of 1.2%
- ▶ Bitcoin has a daily volatility of 3.5%
- ▶ returns are serially uncorrelated with constant variance

▣ Tasks

1. Compute the three simple returns R_t and the three log returns r_t , all in %.
2. Compute the 3-day returns $P_3/P_0 - 1$ and $\ln(P_3/P_0)$ and compare them with $\sum_t R_t$ and $\sum_t r_t$.
3. Annualise the index mean log return and volatility with $P = 252$.
4. Annualise the Bitcoin volatility and justify the choice of P .

▣ **Report:** a table day, price, R_t , r_t and the three annual numbers

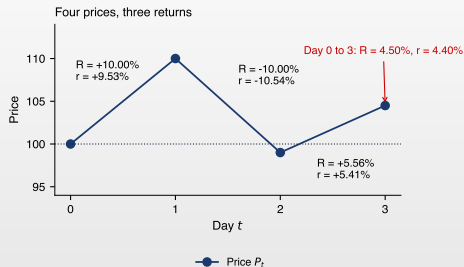
▣ **Notebook:** section A1

A1 [Solved]: Solution

t	P_t	R_t	r_t
0	100	—	—
1	110	+10.00%	+9.53%
2	99	-10.00%	-10.54%
3	104.5	+5.56%	+5.41%
sum		5.56%	4.40%
0 $\rightarrow$ 3		4.50%	4.40%

□ Log returns telescope to $\ln(104.5/100)$

□ Simple returns compound:
 $1.1 \times 0.9 \times 1.0556 - 1 = 4.5\%$



□ Index

- ▶ annualised mean log return $252 \times 0.04\% = 10.08\%$
- ▶ volatility $\sqrt{252} \times 1.2\% = 15.87 \times 1.2\% = 19.05\%$

□ Bitcoin trades every day: $\sqrt{365} \times 3.5\% = 66.9\%$

- ▶ with $\sqrt{252}$ one reports 55.6%, 17% too low

 MFM_ch1_seminar_tests

A4(a) [Proposed]: The Depth of a Drawdown — Task

- ▣ **Model:** A1 [Solved]; notebook section A4
- ▣ **Questions**
 - ▶ what is the worst loss from a previous peak?
 - ▶ how does it compare with the growth rate?
- ▣ **Inputs**
 - ▶ year-end portfolio values $V_0, \dots, V_5 = 100, 120, 90, 110, 130, 117$
 - ▶ 5-year CAGR 3.19%
- ▣ **Tasks:** complete the worksheet
 1. Compute the running peak $M_t = \max_{s \leq t} V_s$.
 2. Compute the drawdown $DD_t = V_t/M_t - 1$, as a negative number.
 3. Find the maximum drawdown $MDD = \min_t DD_t$ and the year in which it occurs.
 4. Compute the Calmar ratio $\text{CAGR}/|MDD|$.

Year	0	1	2	3	4	5
V_t	100	120	90	110	130	117
M_t	100	120	...	...	...	...
DD_t	0%	0%	...	...	...	...

- ▣ **Interpretation:** why is the fall to 117 in year 5 not the maximum drawdown?

B1 [Solved]: Descriptive Statistics — Task

▣ Questions

- ▶ how heavy are the tails of BET-TR, Bitcoin and EUR/RON?
- ▶ how much does the kurtosis depend on a few days?

▣ Data: r_{bet} , r_{btc} , r_{eur} from the setup

- ▶ $T = 2\,992, 4\,384, 5\,352$; $P = 252, 365, 252$

▣ Tasks

1. Compute the mean, the standard deviation, the annualised volatility, the skewness, the excess kurtosis, the minimum and the maximum of each series.
2. Compute $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ and its two terms.
3. Compare JB with 5.99, the 5% critical value of χ^2_2 .
4. Recompute each excess kurtosis after removing the day with the largest $|r_t|$, and after removing the five largest.
5. Recompute the Crow–Siddiqui quantile excess kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$ on the same trimmed samples (0 for the Normal distribution).

▣ Report: one table for steps 1–3 and a chart of kurtosis against the number of days removed

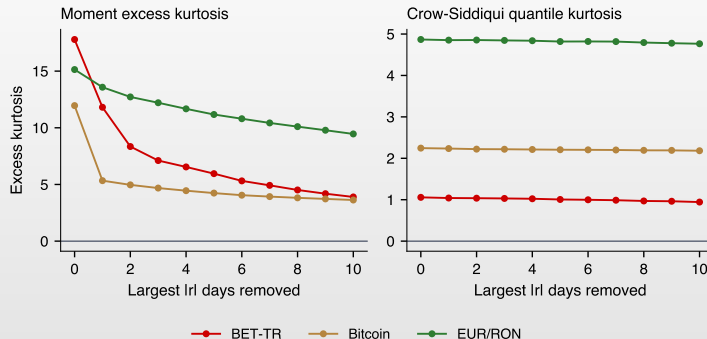
▣ Interpretation: which measure of tail heaviness can you report with confidence?

B1 [Solved]: Solution (1/2) — BET-TR Step by Step

- Step 1: $\bar{r} = 0.077\%$, $\hat{\sigma} = 0.978\%$, volatility $\sqrt{252} \times 0.978\% = 15.5\%$
 - ▶ $\hat{S} = -1.40$, $\hat{K} - 3 = 17.79$
 - ▶ worst day -11.86% on 19 Dec 2018 (the bank-asset tax)
 - ▶ best day $+6.71\%$ on 24 Dec 2018
- Step 2: $JB = 40\,433$
 - ▶ skewness term $2\,992 \times 1.40^2/6 = 977$
 - ▶ kurtosis term $2\,992 \times 17.79^2/24 = 39\,455$, 98% of the statistic
- Step 3: $40\,433 > 5.99$, $p < 10^{-10}$
- Step 4: excess kurtosis 17.8 with all days
 - ▶ 11.8 without 19 Dec 2018
 - ▶ 5.95 without the five largest days: 19 Dec 2018, 16 Mar 2020, 9 Mar 2020, 24 Dec 2018, 9 Mar 2022
- Step 5: Crow–Siddiqui 1.06 with all days, 1.01 without the five largest days
- One day carries a third of the moment kurtosis
 - ▶ the quantile measure does not move

	BET-TR	Bitcoin	EUR/RON
Mean (% a day)	0.077	0.118	0.007
Std. dev. (% a day)	0.978	3.485	0.279
Volatility (% a year)	15.5	66.6	4.4
Skewness	-1.40	-0.70	0.73
Excess kurtosis	17.8	12.0	15.1
Min (%)	-11.9	-46.5	-2.5
Max (%)	6.7	22.5	2.9
JB	40 433	26 482	51 575
Top 5 removed	5.95	4.24	11.17

B1 [Solved]: Solution (2/2) — Sensitivity



- All JB statistics exceed the critical value 5.99 more than 4 000-fold
- Bitcoin without 12 Mar 2020 alone: kurtosis 12.0 → 5.3
- Interpretation: the tails are heavy by both measures
 - ▶ report the quantile excess kurtosis: its value does not hinge on a few days

A3 [Solved]: Jarque–Bera and Ljung–Box, Step by Step — Task

▣ Questions

- ▶ how does a statistic become a decision?
- ▶ which assumptions does the decision need?

▣ Inputs: two hypothetical samples of daily returns; tests at the 5% level

- ▶ sample 1: $T = 2\,500$, skewness $\hat{S} = -0.5$, kurtosis $\hat{K} = 8$ (not excess)
- ▶ sample 2: $T = 1\,000$, $\hat{\rho}_1 = 0.10$, $\hat{\rho}_2 = -0.05$, $\hat{\rho}_3 = 0.03$

▣ Tasks

1. Compute the two terms of $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ Jarque & Bera (1980), the statistic, and $p = P(\chi_2^2 \geq JB)$.
2. Compute the three terms of $Q(3) = T(T+2) \sum_{k=1}^3 \hat{\rho}_k^2 / (T-k)$ Ljung & Box (1978), the statistic, and $p = P(\chi_3^2 \geq Q)$.
3. State the H_0 of each test.
4. Name the assumption of each χ^2 reference distribution that fails under volatility clustering.

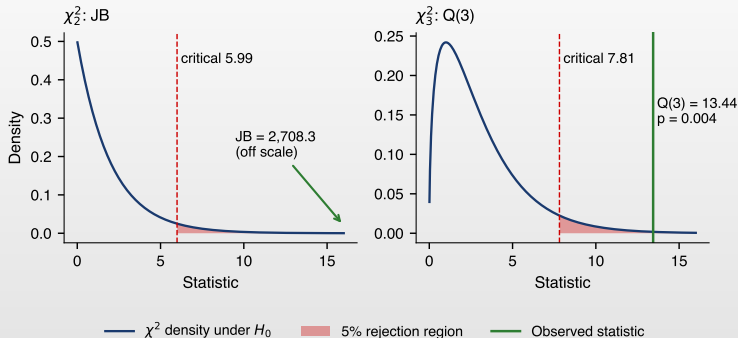
▣ Report: the summands, the two statistics, the p-values and the decisions

▣ Notebook: section A3

A3 [Solved]: Solution (1/2) — Step by Step

1. $\hat{S}^2/6 = 0.25/6 = 0.042$; $(\hat{K} - 3)^2/24 = 25/24 = 1.042$
 - ▶ $JB = 2500 \times 1.084 = 2708.3$, $p < 10^{-10}$: reject
 - ▶ 96% of JB is the kurtosis term
2. $0.01/999 = 1.001 \times 10^{-5}$, $0.0025/998 = 0.251 \times 10^{-5}$, $0.0009/997 = 0.090 \times 10^{-5}$
 - ▶ $Q(3) = 1000 \times 1002 \times 1.342 \times 10^{-5} = 13.44$, $p = 0.004$: reject
3. H_0 of each test
 - ▶ JB: i.i.d. returns with the Normal distribution
 - ▶ Ljung–Box: no autocorrelation
4. The χ^2_3 reference of Ljung–Box also needs i.i.d. returns
 - ▶ under clustering $\text{Var}(\hat{\rho}_k) > 1/T$ and Q rejects too often (A10, B9)

A3 [Solved]: Solution (2/2) — Reference Distributions



- The p-value is the area of the χ^2 density beyond the observed statistic
- Interpretation: for an i.i.d. sample both rejections are sound
 - ▶ for clustered returns only the JB rejection of "i.i.d. and Normal" is sound
 - ▶ the Q(3) p-value is then too small

B2 [Proposed]: Heavy Tails and VaR 1% on BET-TR — Task (1/2)

- **Model:** B1 [Solved]; notebook section B2
- **Question:** how large is the one-day loss exceeded on 1% of days, under three distributions?
- **Data:** r_{bet} , $T = 2\,992$
 - ▶ models fitted and checked on the **same** sample: a descriptive comparison
 - ▶ not a backtest (a check of VaR on data after the estimation window)
- **Tasks**
 1. Standardise $z_t = (r_t - \bar{r})/\hat{\sigma}$ and count the days with $|z_t| > 3$.
 2. Compare their share with 0.27% under the Normal distribution.
 3. Fit a Student- t to z_t by maximum likelihood (`scipy.stats.t.fit`: degrees of freedom ν , location m , scale s).

B2 [Proposed]: Heavy Tails and VaR 1% on BET-TR — Task (2/2)

▣ **Tasks** (continued)

4. Compute $\text{VaR}_{1\%} = -q_r$, with $q_r = \bar{r} + \hat{\sigma}q_z$, for three choices of the 1% quantile.

- Student-t: $q_z = m + s t_{\nu}^{-1}(0.01)$
- Normal distribution: $q_z = \Phi^{-1}(0.01) = -2.326$
- empirical: $q_r =$ the 1% quantile of r_t

5. Count the breaches ($r_t < q_r$) of each VaR and compare them with $0.01 T = 29.9$.

▣ **Report:** a table method / VaR / breaches and a lower-tail plot with the three quantiles

▣ **Interpretation:** why does the empirical VaR have about 30 breaches by construction?

A6 [Solved]: Kurtosis of a Volatility Mixture — Task

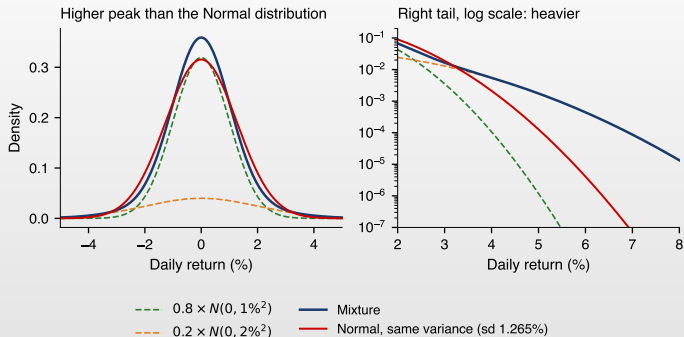
- ▣ **Question:** can random volatility alone create heavy tails?
- ▣ **Assumptions:** $r_t = \sigma_t \varepsilon_t$, ε_t i.i.d. $N(0, 1)$ and independent of σ_t
 - ▶ $\sigma_t = 1\%$ with probability 0.8 and 2% with probability 0.2
 - ▶ σ_t is drawn **independently every day**
- ▣ **Tasks**
 1. Using independence and $\mathbb{E}\varepsilon^4 = 3$, show that $K = \mathbb{E}r^4 / (\mathbb{E}r^2)^2 = 3\mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2$.
 2. Complete a two-state table with the probability, σ^2 (in $\%^2$) and σ^4 (in $\%^4$).
 3. Compute $\mathbb{E}\sigma^2$, $\mathbb{E}\sigma^4$ and K .
 4. Use Jensen's inequality, $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$ with $X = \sigma^2$, to show that $K \geq 3$.
 5. State whether this model produces volatility clustering.
- ▣ **Report:** the table, K and the excess kurtosis
- ▣ **Interpretation:** link the result to the mixture-of-distributions hypothesis of [Clark \(1973\)](#)
- ▣ Notebook: section A6

A6 [Solved]: Solution (1/2) — Moments

State	Prob.	σ^2	σ^4
calm	0.8	1	1
turbulent	0.2	4	16
mean		1.6	4.0

1. $\mathbb{E}r^4 = \mathbb{E}\sigma^4 \mathbb{E}\varepsilon^4 = 3\mathbb{E}\sigma^4$ and $\mathbb{E}r^2 = \mathbb{E}\sigma^2$
2. The table on the left
3. $\mathbb{E}\sigma^2 = 1.6$, $\mathbb{E}\sigma^4 = 4.0$
 - $K = 3 \times 4.0 / 1.6^2 = 12 / 2.56 = 4.69$; excess kurtosis 1.69
4. $\mathbb{E}[(\sigma^2)^2] \geq (\mathbb{E}\sigma^2)^2$, with equality only for a constant σ : $K \geq 3$
5. No clustering: the states are independent across days
 - so $|r_t|$ is uncorrelated
 - A8 makes the states persistent

A6 [Solved]: Solution (2/2) — Densities

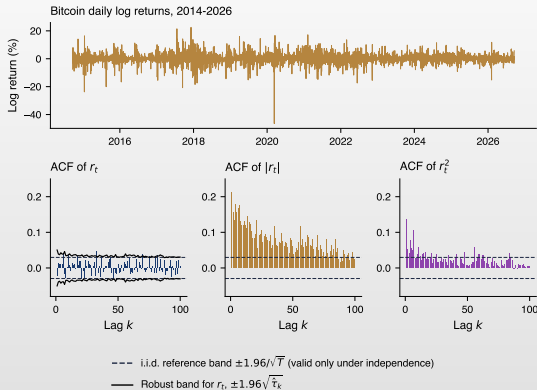


- $P(|r_t| > 4\%)$: 0.92% in the mixture, 0.16% for the Normal distribution with the same variance
- Interpretation: the random operational time of [Clark \(1973\)](#) (the pace of information arrival) is such a mixture
 - variation in volatility creates heavy tails

B3 [Solved]: Uncorrelated but Not Independent (Bitcoin) — Task

- ▣ **Question:** do past Bitcoin returns predict the direction or only the size of today's return?
- ▣ **Data:** r_{btc} , 18 Sep 2014 – 18 Sep 2026, $T = 4\,384$
- ▣ **Tasks**
 1. Plot the returns.
 2. Plot the ACF of r_t , $|r_t|$ and r_t^2 for lags 1–100 with the i.i.d. reference band $\pm 1.96/\sqrt{T} = \pm 0.030$.
 3. Compute the Ljung–Box $Q(10)$ on r_t and on r_t^2 (acorr_ljungbox), with p-values from χ_{10}^2 .
 4. Run the ARCH-LM test with 5 lags [Engle \(1982\)](#).
 - regress x_t^2 on a constant and $x_{t-1}^2, \dots, x_{t-5}^2$
 - compare $(T - 5)R^2$ with χ_5^2
- ▣ **Report:** the chart, the lag-1 autocorrelations and the three statistics
 - ▶ label each with its series and its H_0
- ▣ **Interpretation:** the $Q(10)$ on r_t has $p \approx 3\%$
 - ▶ given step 4, is its i.i.d. reference distribution valid?
- ▣ Notebook: section B3

B3 [Solved]: Solution (1/2) — Returns and ACF



- Direction: no structure in the ACF of r_t
- Size: $|r_t|$ and r_t^2 autocorrelated for dozens of lags

B3 [Solved]: Solution (2/2) — Tests

2. $\hat{\rho}_1$: -0.022 for r_t (inside the band), 0.213 for $|r_t|$ and 0.137 for r_t^2
3. $Q(10)$ results
 - ▶ on r_t : 20.2 , $p = 0.028$
 - ▶ on r_t^2 : 212.7 , $p \approx 4 \times 10^{-40}$
4. $\text{ARCH-LM}(5) = 114.7$, $p \approx 4 \times 10^{-23}$
 - ▶ its χ^2 reference needs $\mathbb{E}r^4 < \infty$, so the p-value is indicative
- The sizes of returns are dependent: returns are not i.i.d.
 - ▶ the 2.8% p-value of $Q(10)$ on r_t rests on a false assumption
- The i.i.d. band is a reference, not a valid test for $|r_t|$ and r_t^2
 - ▶ A10 and B9 correct the inference for r_t

B9 [Solved]: Heteroskedasticity-Robust Ljung–Box — Task (1/2)

- ▣ **Question:** which autocorrelations of returns survive once the variance of $\hat{\rho}_k$ allows for clustering?
- ▣ **Data:** $r_{\text{sp}}, r_{\text{bet}}, r_{\text{eur}}, r_{\text{btc}}$; $x_t = r_t - \bar{r}$
- ▣ **Tasks**
 1. For $k = 1, \dots, 20$ compute $\hat{\rho}_k$ and $\hat{\tau}_k = \sum_{t=k+1}^T x_t^2 x_{t-k}^2 / (\sum_{t=1}^T x_t^2)^2$ [Lobato, Nankervis & Savin \(2001\)](#).
 2. Plot $\hat{\rho}_k$ with the classical band $\pm 1.96/\sqrt{T}$ and the robust band $\pm 1.96\sqrt{\hat{\tau}_k}$.
 3. Compute the classical $Q(10)$ and the robust $\tilde{Q}(10) = \sum_{k=1}^{10} \hat{\rho}_k^2 / \hat{\tau}_k$.
 4. Compare both with χ_{10}^2 , whose 5% critical value is 18.3.
 5. Count the lags among 1–20 outside each band.
 - these are pointwise checks, distinct from the joint 10-lag test of steps 3–4

B9 [Solved]: Heteroskedasticity-Robust Ljung–Box — Task (2/2)

▣ Assumptions of $\tilde{Q}$

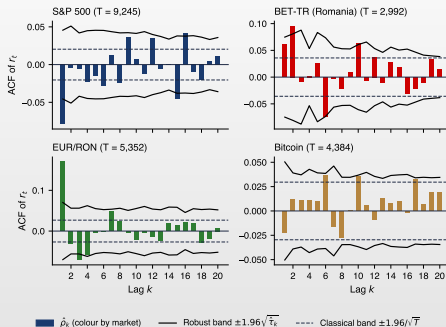
- ▶ a finite fourth moment, $\mathbb{E}r^4 < \infty$
- ▶ zero cross terms, $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for $k \neq l$

▣ Interpretation

- ▶ for which series does autocorrelation survive?
- ▶ is the fourth-moment assumption credible?

▣ Notebook: section B9

B9 [Solved]: Solution (1/3) — ACF with Both Bands



- S&P 500, lag 1: $\hat{\rho}_1 = -0.079$, $T\hat{\tau}_1 = 5.00$
 - the robust band is $1.96\sqrt{5.00/9245} = 0.046$ instead of 0.020
- At lag 1 the robust band is $\sqrt{T\hat{\tau}_1} = 1.7$ (Bitcoin) to 2.6 (EUR/RON) times wider than the classical one

B9 [Solved]: Solution (2/3) — Joint Tests and Interpretation

	S&P 500	BET-TR	EUR/RON	Bitcoin
$Q(10)$ (p)	91.6 ($< 10^{-10}$)	70.0 ($< 10^{-10}$)	221.7 ($< 10^{-10}$)	20.2 (0.028)
$\tilde{Q}(10)$ (p)	19.0 (0.041)	16.9 (0.076)	37.4 (< 0.001)	11.9 (0.29)
Lags outside, classical / robust	8 / 3	5 / 2	6 / 2	3 / 0
Lags outside the robust band	1, 15, 16	2, 10	1, 3	none

- At 5% the classical test rejects for all four series
 - ▶ the robust test rejects for the S&P 500 (barely) and EUR/RON only
- Bitcoin (B3): the 2.8% p-value becomes 29%
 - ▶ no evidence of linear autocorrelation at lags 1–10
- EUR/RON keeps $\hat{\rho}_1 = 0.17$
 - ▶ under the managed float, the BNR reference rate adjusts in small steps over several days

B9 [Solved]: Solution (3/3) — Remaining Assumptions

- Zero cross terms
 - ▶ the test of [Dalla, Giraitis & Phillips \(2022\)](#) keeps only the significant cross terms
 - ▶ it finds none for the S&P 500, BET-TR and Bitcoin
- Finite fourth moment
 - ▶ if $P(|r_t| > x)$ decays like $x^{-\nu}$ with $\nu \leq 4$, $\mathbb{E}r^4$ is infinite
 - ▶ the robust p-values are then not valid
 - ▶ A7 and B11 estimate this exponent after the lecture

B4 [Proposed]: Aggregational Gaussianity (Bitcoin) — Task

- ▣ **Model:** B1 and B3 [Solved]; notebook section B4
- ▣ **Question:** do h -day Bitcoin returns approach the Normal distribution as h grows?
- ▣ **Data:** r_btc , $T = 4\,384$, in non-overlapping blocks from the first day
 - ▶ the first 5-day block is 18–22 Sep 2014, the second 23–27 Sep 2014
 - ▶ the incomplete last block (4 days for $h = 5$ and $h = 20$) is dropped
- ▣ **Tasks**
 1. Build the h -day log returns (sums of h consecutive daily returns) for $h = 1, 5, 20$.
 2. Draw a QQ plot of each standardised series against the Normal distribution, with N , the number of blocks, in the title.
 3. Compute the excess kurtosis and the i.i.d.-Gaussian reference SE $\sqrt{24/N}$.
- ▣ **Report:** the three QQ plots and a table h , N , excess kurtosis, $\sqrt{24/N}$
- ▣ **Interpretation**
 - ▶ why is $\sqrt{24/N}$ too optimistic here?
 - ▶ why do non-overlapping blocks remove shared days but not volatility dependence?

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “Are the daily BET-TR log returns Normally distributed? Are they autocorrelated? Give Python code and a conclusion”
- **Answer:** the code below and the conclusion

```
from scipy import stats
from statsmodels.stats.diagnostic import acorr_ljungbox
r = np.log(bet["close"]).diff().dropna()      # daily log returns
k = stats.kurtosis(r)                          # sample kurtosis
print("excess_kurtosis:", k - 3)              # the Normal has kurtosis 3
jb, p = stats.jarque_bera(r)
lb = acorr_ljungbox(r**2, lags=[10])          # Ljung-Box, 10 lags
```

- **Conclusion of the AI**
 - ▶ “Excess kurtosis 14.8: heavy tails”
 - ▶ “JB p -value 0.000: the probability that BET-TR returns follow the Normal distribution is 0%”
 - ▶ “Ljung-Box $Q(10) = 713.5$, $p < 0.001$: returns are strongly autocorrelated, so BET-TR is predictable and not a random walk”

C2 [Proposed]: Critique an AI Answer — Task (1/2)

- ▣ **Model:** A3, B3 and B9 [Solved]; notebook section C2
- ▣ **Data:** `r_bet`, the 2992 returns of the setup
- ▣ **Tasks**
 1. Find every substantive error in the code and in the three conclusions.
 2. For each error, say how you would detect it.
 - a test on simulated data or a number from the primer slides
 - a solved exercise or the library documentation
 3. Write corrected, executable code.

C2 [Proposed]: Critique an AI Answer — Task (2/2)

▣ **Tasks** (continued)

4. Report the excess kurtosis, JB with its p-value, $Q(10)$ on r_t and on r_t^2 , and the robust $\tilde{Q}(10)$ of B9.
5. State the assumptions behind the robust p-value.
6. Write the corrected conclusion in three sentences.

▣ **Report**

- ▶ the corrected code
 - ▶ a table AI value / correct value
 - ▶ a chart of the ACF of r_t and of r_t^2
- ▣ AI code usually runs without an error message
- ▶ the mistakes are in the statistics, not in the syntax
 - ▶ log what you find in `AI_ERRORS.md`

A7 [Solved]: The Hill Estimator, Step by Step — Task

▣ Questions

- ▶ how is a tail index estimated from the largest observations?
- ▶ how precise is it?

▣ **Inputs:** the 11 largest absolute daily returns (%) of a hypothetical sample: 12.0; 9.5; 8.2; 7.4; 6.9; 6.1; 5.8; 5.5; 5.2; 5.0; 4.8

▣ **Assumption:** above the threshold $X_{(k+1)}$ the tail is Pareto, $P(X > x) \propto x^{-\alpha}$

- ▶ the observations are independent

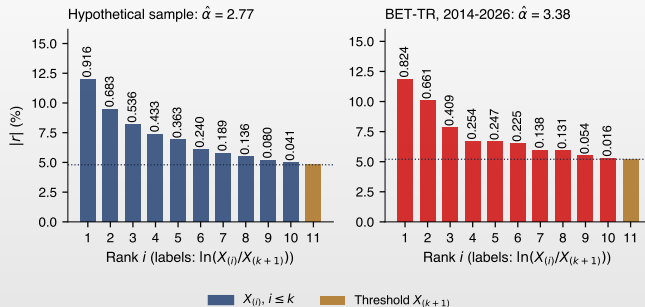
▣ Tasks

1. For $k = 10$ compute the log-ratios $\ln(X_{(i)}/X_{(11)})$, their mean and $\hat{\alpha} = 1/\text{mean}$ Hill (1975).
2. Compute $\text{SE} = \hat{\alpha}/\sqrt{k}$ and the interval $\hat{\alpha} \pm 1.96 \text{ SE}$.
3. State which moments $\mathbb{E}|X|^p$ ($p < \alpha$) exist at the point estimate.
4. State which of these claims hold at 95%.
5. Repeat steps 1–2 on the 11 largest $|r_t|$ of BET-TR, using the worksheet printed by the notebook.

▣ **Report:** $\hat{\alpha}$, SE and interval for both samples

▣ Notebook: section A7

A7 [Solved]: Solution (1/2) — Hypothetical Sample



1. The labels are the log-ratios to 4.8: sum 3.616, mean 0.3616, $\hat{\alpha} = 1/0.3616 = 2.77$
2. $SE = 2.77/\sqrt{10} = 0.87$; interval [1.05; 4.48]
 ► a rough approximation by the Normal distribution, valid for i.i.d. Pareto tails
3. At 2.77 the variance exists and the fourth moment does not
4. The interval contains values below 2 and above 4: neither claim holds at 95%

A7 [Solved]: Solution (2/2) — BET-TR Worksheet

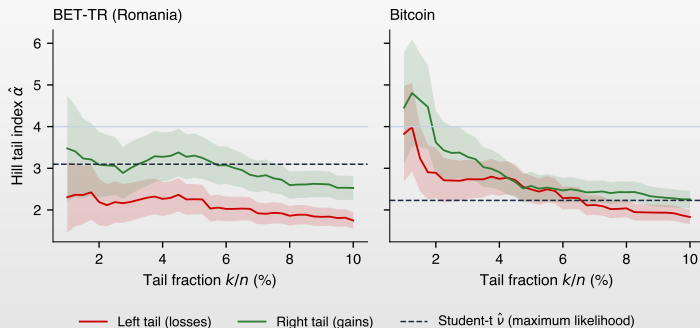
i	Date	r_t (%)	$X_{(i)}$ (%)	$\ln(X_{(i)}/X_{(11)})$
1	19 Dec 2018	-11.857	11.857	0.824
2	16 Mar 2020	-10.075	10.075	0.661
3	9 Mar 2020	-7.827	7.827	0.409
4	24 Dec 2018	+6.707	6.707	0.254
5	9 Mar 2022	+6.662	6.662	0.247
6	24 Aug 2015	-6.512	6.512	0.225
7	24 Mar 2020	+5.972	5.972	0.138
8	17 Mar 2020	+5.928	5.928	0.131
9	4 Mar 2022	-5.490	5.490	0.054
10	12 Mar 2020	-5.285	5.285	0.016
11	18 Jan 2016	-5.202	5.202	threshold

- Row 1 worked: $\ln(11.857/5.202) = \ln 2.279 = 0.824$
- Sum of the ten log-ratios 2.958;
 $\hat{\alpha}_{10} = 10/2.958 = 3.38$
- $SE = 3.38/\sqrt{10} = 1.07$; interval [1.28; 5.48]
- Seven of the ten days fall in December 2018 or March 2020
 - ▶ the extremes cluster, so the i.i.d. SE understates the uncertainty
- Interpretation: ten observations cannot decide whether the fourth moment exists
 - ▶ B11 uses $k = 2\%$ of n

B11 [Solved]: Hill on Both Tails, BET-TR and Bitcoin — Task

- ▣ **Question:** are large losses more frequent than large gains of the same size?
- ▣ **Data:** r_{bet} ($n = 2992$), r_{btc} ($n = 4384$)
 - ▶ losses $\text{loss} = -r[r < 0]$, gains $\text{gain} = r[r > 0]$
 - ▶ each sorted in decreasing order
- ▣ **Tasks**
 1. Plot the Hill estimate $\hat{\alpha}(k)$ of each tail for k from 1% to 10% of the full-sample n , with the band $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$.
 2. Compute $\hat{\alpha}_-$ (losses) and $\hat{\alpha}_+$ (gains) at $k = \lfloor 0.02 n \rfloor$ (BET-TR 59, Bitcoin 87), with 95% intervals.
 - the lecture uses $k = 2.5\%$ of n for $|r_t|$, which pools both tails
 3. Compute the approximate independent-tail statistic $z = (\hat{\alpha}_- - \hat{\alpha}_+)/\sqrt{\hat{\alpha}_-^2/k + \hat{\alpha}_+^2/k}$ and its two-sided p-value.
 4. Compare with the Student- t $\hat{\nu}$ fitted to the standardised returns: 3.10 for BET-TR (B2), 2.23 for Bitcoin.
- ▣ **Interpretation**
 - ▶ is the gain/loss asymmetry statistically established?
 - ▶ does clustering make the true SE larger or smaller?
- ▣ Notebook: section B11

B11 [Solved]: Solution (1/2) — Hill Plots and Estimates



- BET-TR: $\hat{\alpha}_- = 2.19$ [1.63; 2.75], $\hat{\alpha}_+ = 3.08$ [2.30; 3.87]
- Bitcoin: $\hat{\alpha}_- = 2.89$ [2.28; 3.50], $\hat{\alpha}_+ = 3.62$ [2.86; 4.38]
- E.g. BET-TR losses: $2.19 \times (1 \pm 1.96/\sqrt{59}) = 2.19 \times (1 \pm 0.255)$

B11 [Solved]: Solution (2/2) — Test and Interpretation

- BET-TR: $\sqrt{2.19^2/59 + 3.08^2/59} = \sqrt{0.0812 + 0.1612} = 0.492$; $z = (2.19 - 3.08)/0.492 = -1.82$, $p = 0.069$
- Bitcoin: $\sqrt{2.89^2/87 + 3.62^2/87} = 0.497$; $z = (2.89 - 3.62)/0.497 = -1.47$, $p = 0.14$
- Decision: losses are heavier in both markets, but $H_0: \alpha_- = \alpha_+$ is not rejected at 5%
 - ▶ the statistic treats the two tails as independent samples
 - ▶ extremes cluster in time, so the true SE is larger and the evidence weaker
 - ▶ a joint block bootstrap of the original returns keeps both features (extension)
- $\hat{\nu}$ is fitted to the whole distribution, while $\hat{\alpha}$ uses only the tail
 - ▶ Bitcoin's 2.23 lies below both tail indices
 - ▶ the Student- t does not fit centre and tails together

A10 [Proposed]: Widening of the Autocorrelation Band under Clustering — Task (1/2)

- **Model:** A6 [Solved]; notebook section A10
- **Setting:** $x_t = \sigma_t \varepsilon_t$ with ε_t i.i.d. $N(0, 1)$, independent of the stationary volatility process
 - ▶ then x_t is uncorrelated
 - ▶ $\text{Var}(\hat{\rho}_1) \approx v_1/T$, with $v_1 = \mathbb{E}[x_t^2 x_{t-1}^2] / (\mathbb{E}x_t^2)^2$ [Lobato, Nankervis & Savin \(2001\)](#)
- **Inputs**
 - ▶ A6: $\mathbb{E}\sigma^2 = 1.6$, $\text{Var}(\sigma^2) = 4.0 - 1.6^2 = 1.44$ (in %⁴)
 - ▶ persistent regimes (A8): $\text{Cov}(\sigma_t^2, \sigma_{t-1}^2) = \lambda \text{Var}(\sigma^2)$ with $\lambda = 0.875$
 - ▶ S&P 500: $T = 9\,245$, $\hat{\rho}_1 = -0.079$, $T\hat{\tau}_1 = 5.00$

A10 [Proposed]: Widening of the Autocorrelation Band under Clustering — Task (2/2)

▣ Tasks

1. Show that $v_1 = 1$ for the A6 mixture, whose states are independent across days.
2. Show that $v_1 = 1 + \lambda \text{Var}(\sigma^2)/(\mathbb{E}\sigma^2)^2$ with persistent regimes.
3. Compute v_1 and the band-widening factor $\sqrt{v_1}$.
4. Compute the S&P 500 lag-1 t -statistic with the i.i.d. SE $1/\sqrt{T}$ and with the robust SE $\sqrt{\hat{\tau}_1}$.
5. Compute the two 95% band half-widths.

▣ **Interpretation:** which feature widens the band: heavy tails or persistent volatility?

After the Lecture: Remaining Tasks

- These three tasks use tools that [Chapter 1](#) introduces
 - ▶ A7 [Solved] and B11 [Solved]: the Hill estimator of the tail index, step by step and on both tails
 - ▶ A10 [Proposed]: volatility clustering, not heavy tails, widens the autocorrelation band of B9
- Work on them after the lecture, before the next seminar
- The solution of A10 is discussed at the next seminar

Extension Packs: Content and Prerequisites

Pack	Exercises	Prerequisites
1. Return arithmetic and drawdowns	A1(c) [Solved], A2, A4(b–e)	A1, A4(a)
2. Mixtures, regimes and tails	A5 [Solved], A8, A9, A11	A6, A7, A10
3. Robust moments and model checks	B1(d) [Solved], B10, B14	B1, B9, B11
4. Dependence and asymmetry	B5, B12, B6, B15	B3, B9
5. Performance and range estimators	B7, B13, B8	A1, A5, B1
6. Projects	C1, C3	B9, the lecture case study

- Notebook sections carry the same labels as the exercises
- Solutions of [Proposed] exercises are discussed at the next seminar

A1(c) [Solved]: Portfolio Returns — Task and Solution

- ▣ **Question:** which return adds up across assets?
- ▣ **Inputs:** 60% in asset A and 40% in asset B for one day; $R_A = +5\%$, $R_B = -2\%$
- ▣ **Tasks**
 1. Compute the portfolio simple return $R_p = 0.6R_A + 0.4R_B$.
 2. Compute the weighted log return $0.6r_A + 0.4r_B$.
 3. Compute the true portfolio log return $\ln(1 + R_p)$.
- ▣ **Solution**
 1. $R_p = 0.6 \times 5\% + 0.4 \times (-2\%) = 2.20\%$
 2. $0.6 \times \ln 1.05 + 0.4 \times \ln 0.98 = 0.6 \times 4.879\% - 0.4 \times 2.020\% = 2.119\%$
 3. $\ln 1.022 = 2.176\%$
- ▣ Interpretation: simple returns add across assets, log returns add over time (A1)
 - ▶ for daily moves the gap between the two rules is small
 - ▶ for large moves it is not

A2 [Proposed]: The $\sqrt{h}$ Rule under Autocorrelation — Task

- ▣ **Model:** A1 [Solved]; notebook section A2
- ▣ **Assumption:** stationary AR(1), $r_t = \rho r_{t-1} + e_t$, $|\rho| < 1$, $\text{Var}(r_t) = \sigma^2$, so $\gamma_j = \text{Cov}(r_t, r_{t-j}) = \sigma^2 \rho^j$
- ▣ **Scaffold**, $h = 3$: $\text{Var}(r_1 + r_2 + r_3) = 3\gamma_0 + 2(2\gamma_1 + \gamma_2)$: two pairs at lag 1, one pair at lag 2
- ▣ **Tasks**
 1. Generalise the pair count: show that $\text{Var}(\sum_{j=1}^h r_{t+j}) = h\sigma^2[1 + 2\sum_{j=1}^{h-1}(1 - j/h)\rho^j]$.
 2. Using $\sum_{j=1}^{h-1} \rho^j = \rho(1 - \rho^{h-1})/(1 - \rho)$ and $\sum_{j \geq 1} j|\rho|^j = |\rho|/(1 - |\rho|)^2$, show that the variance ratio $VR(h) = \text{Var}(\cdot)/(h\sigma^2) \rightarrow (1 + \rho)/(1 - \rho)$.
 3. Illustrative calculation: compute $\sqrt{VR(252)}$ and the corrected annual volatilities, treating the sample lag-1 autocorrelations as AR(1) parameters.
 - EUR/RON: $\rho = 0.170$, volatility 4.44%
 - S&P 500: $\rho = -0.079$, volatility 17.99%
 4. Explain why volatility clustering alone does not change step 1.
- ▣ **Interpretation:** which ACF check must precede step 3?

A4(b–e) [Proposed]: A Benchmark for Drawdowns — Task (1/2)

- ▣ **Model:** A4(a); notebook section A4
- ▣ **Assumption:** the log price is $X_t = \mu t + \sigma W_t$
 - ▶ W_t a standard Brownian motion, σ the annual volatility, T the horizon in years
 - ▶ maximum log drawdown $D = \max_{s \leq t \leq T} (X_s - X_t) = -\ln(1 + MDD)$
 - ▶ for $\mu = 0$: $\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ Magdon-Ismail et al. (2004)
- ▣ **Inputs:** gold spot, 1990–2026
 - ▶ $\sigma = 16.3\%$, $T = 36.7$ years, mean log return 6.5% a year
 - ▶ observed $MDD = -44.7\%$, so $D = 0.592$

A4(b–e) [Proposed]: A Benchmark for Drawdowns — Task (2/2)

▣ Tasks

1. Use $W(cT) \stackrel{d}{=} \sqrt{c} W(T)$ to explain why $\mathbb{E}[D]$ is proportional to $\sqrt{T}$ when $\mu = 0$.
2. Compute the ratio of $\mathbb{E}[D]$ for 40 and 10 years of the same process.
3. Compute the driftless benchmark for gold and compare it with the observed D .
4. Simulate 2000 daily paths (seed 42) with $\mu = 0$ and with $\mu = 6.5\%$.
5. Plot the mean D against T for both drifts.

- ▣ **Interpretation:** what do steps 1–5 imply for Calmar ratios computed on samples of different length?

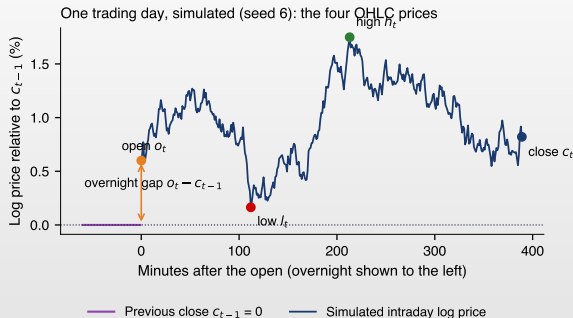
A5 [Solved]: Parkinson Volatility and Its Efficiency — Task

- ▣ **Question:** how much information does the daily range carry about volatility?
- ▣ **Notation:** o_t, h_t, l_t, c_t = log open, high, low, close; $h_t - l_t = \ln(H_t/L_t)$
- ▣ **Inputs:** H/L of five days: 1.020, 1.015, 1.030, 1.010, 1.025; $P = 252$
- ▣ **Assumption:** within the day the log price is a driftless Brownian motion with volatility σ
 - ▶ no overnight jumps
- ▣ **Tasks**
 1. Complete a five-row table: $\ln(H/L)$, its square, and the mean of the squares.
 2. Compute $\hat{\sigma}_P^2 = \overline{(h - l)^2} / (4 \ln 2)$ [Parkinson \(1980\)](#), the daily and the annual volatility.
 3. Name the information the estimator ignores.
 4. Explain the effect of a strong drift on the estimate.
 5. Using $\mathbb{E}[(h - l)^2] = 4 \ln 2 \sigma^2$ and $\mathbb{E}[(h - l)^4] = 9\zeta(3) \sigma^4$ ($\zeta(3) = 1.202$), compare the per-day variance of $(h - l)^2 / (4 \ln 2)$ with $\text{Var}(r_t^2) = 2\sigma^4$.
- ▣ Notebook: section A5

A5 [Solved]: Solution (1/2) — The Estimate

H/L	$\ln(H/L)$	square ($\times 10^{-4}$)
1.020	0.0198	3.92
1.015	0.0149	2.22
1.030	0.0296	8.74
1.010	0.0100	0.99
1.025	0.0247	6.10
mean		4.39

- $\hat{\sigma}_P^2 = 0.000439/2.7726 = 0.000158$; $\hat{\sigma}_P = 1.26\%$ a day, $1.26\% \times 15.87 = 20.0\%$ a year
- Ignored: the open and the overnight jump Yang & Zhang (2000)
- A drift widens the range and biases $\hat{\sigma}_P$ up Rogers & Satchell (1991)



- A simulated day: the range $h_t - l_t$ uses the whole path, the close-to-close return only its end points

A5 [Solved]: Solution (2/2) — Efficiency

5. $\text{Var}((h - l)^2) = (9\zeta(3) - 16(\ln 2)^2)\sigma^4 = (10.819 - 7.687)\sigma^4 = 3.131\sigma^4$

- ▶ divided by $(4 \ln 2)^2 = 7.687$: $0.407\sigma^4$ per day for the Parkinson term
- ▶ close-to-close: $\text{Var}(r_t^2) = 3\sigma^4 - \sigma^4 = 2\sigma^4$ per day
- For an n -day average of independent days, both variances are divided by n
 - ▶ the ratio stays $2/0.407 = 4.9$
- One day of ranges carries about the information of five days of close-to-close returns
- Interpretation: the gain assumes driftless continuous trading without overnight jumps
 - ▶ a stale open (an open computed from prices of components that have not yet traded) affects the estimators that use the open
 - ▶ namely Garman–Klass, Rogers–Satchell and Yang–Zhang, not Parkinson (B8)

A8 [Proposed]: The Origin of the Autocorrelation of r_t^2 — Task (1/2)

- ▣ **Model:** A6 [Solved] and A10; notebook section A8
- ▣ **Assumption:** as in A6, $\sigma_t^2 \in \{1, 4\}$ (in %²) with stationary probabilities (0.8; 0.2)
 - ▶ but the state follows a two-state Markov chain [Hamilton \(1989\)](#)
 - ▶ transition matrix $\begin{pmatrix} 0.975 & 0.025 \\ 0.1 & 0.9 \end{pmatrix}$ (rows: today calm L , turbulent H)
 - ▶ $p_{LL} = 0.975$ and $p_{HH} = 0.9$ are the probabilities of staying in the same state
 - ▶ $\pi_H = 0.2$
- ▣ **Tasks**
 1. Check the balance condition $0.8 \times 0.025 = 0.2 \times 0.1$.
 2. Compute $\lambda = p_{LL} + p_{HH} - 1$.

A8 [Proposed]: The Origin of the Autocorrelation of r_t^2 — Task (2/2)

▣ **Tasks** (continued)

3. With the indicator $I_t = 1$ in the turbulent state, $\sigma_t^2 = 1 + 3I_t$; show that $\text{Cov}(I_t, I_{t-k}) = \pi_H(1 - \pi_H)\lambda^k$.
4. Deduce that $\text{Cov}(r_t^2, r_{t-k}^2) = \text{Var}(\sigma^2)\lambda^k$ for $k \geq 1$.
5. Using A6, compute $\text{Var}(\sigma^2)$, $\text{Var}(r^2) = 3\mathbb{E}\sigma^4 - (\mathbb{E}\sigma^2)^2$ and $\text{corr}(r_t^2, r_{t-k}^2)$ at lags 1 and 10.
6. Plot the implied ACF against the S&P 500 ACF of r_t^2 (0.31 at lag 1, 0.04 at lag 100).

▣ **Interpretation:** which two features of the data can this model not reproduce?

A9 [Proposed]: The S&P 500 Tails in a GARCH with Normal Innovations — Task (1/2)

- **Model:** A6 [Solved]; notebook section A9
 - ▶ an advanced preview of [Chapter 5](#)
- **Model equations** [Bollerslev \(1986\)](#): $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, z_t i.i.d. $N(0, 1)$
 - ▶ $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$
 - ▶ here α is a GARCH coefficient, not a tail index
- **Result:** if $1 - (\alpha + \beta)^2 - 2\alpha^2 > 0$, then $K = 3[1 - (\alpha + \beta)^2]/[1 - (\alpha + \beta)^2 - 2\alpha^2]$

A9 [Proposed]: The S&P 500 Tails in a GARCH with Normal Innovations — Task (2/2)

▣ Tasks

1. Compute K for $(\alpha, \beta) = (0.10; 0.85)$ and $(0.10; 0.88)$.
2. State the condition for a finite fourth moment.
3. Compare it with $\alpha + \beta < 1$.
4. With $\alpha + \beta = 0.99$, solve for the α that gives the S&P 500 kurtosis $K = 13.9$.
5. Fit the model to `r_sp` in %.
 - `arch_model(100*r_sp, mean='Constant', vol='GARCH', p=1, q=1, dist='normal')`
6. Compute the implied K .
7. Draw a QQ plot of $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$.

▣ **Interpretation:** what does the QQ plot say about the Normal distribution of z_t ?

A11 [Proposed]: The Hill Estimator as a Maximum Likelihood Estimator — Task

- ▣ **Model:** A7 [Solved]; notebook section A11
- ▣ **Assumption:** given the threshold $u = X_{(k+1)}$, the k exceedances are the order statistics of an i.i.d. Pareto sample
 - ▶ Pareto density $f(x) = \alpha u^\alpha x^{-\alpha-1}$, $x > u$
- ▣ **Tasks**
 1. Write the log-likelihood $\ell(\alpha)$ of the k exceedances (the ordering adds only the constant $\ln k!$).
 2. Solve $\ell'(\alpha) = 0$ and show that the maximiser is the Hill estimator $\hat{\alpha} = k / \sum_{i=1}^k \ln(X_{(i)}/u)$ [Hill \(1975\)](#).
 3. Compute $\ell''(\alpha)$, the information k/α^2 and $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$.
 4. Check the SE on the data of A7.
 5. Simulate $n = 2\,000\,000$ values $|X|$, X Student- t with $\nu = 3$ (`numpy.random.default_rng(1)`).
 6. Plot $\hat{\alpha}_k$ against k/n and report $k = 200, 2\,000, 20\,000, 200\,000$.
- ▣ **Interpretation:** why does $\hat{\alpha}_k$ drift away from 3 when k is large?

B1(d) [Solved]: Quantile-Based Moments with Bootstrap Intervals (1/2)

- ▣ **Question:** do negative skewness and excess kurtosis survive measures that need no moments?
- ▣ **Measures** Kim & White (2004), both 0 for the Normal distribution
 - ▶ Hinkley skewness $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$
 - ▶ Crow–Siddiqui kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$
- ▣ **Bootstrap:** 1000 resamples (`default_rng(42)`), 95% percentile interval
 - ▶ draw $\lceil T/20 \rceil$ starting days uniformly and take 20 consecutive returns from each
 - ▶ cut to T and recompute both measures

B1(d) [Solved]: Quantile-Based Moments with Bootstrap Intervals (2/2)

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui [95% CI]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]

- Every kurtosis interval lies above 0: heavy tails are established without moments
- Moment skewness and Hinkley skewness measure different things
 - ▶ every Hinkley interval contains 0
 - ▶ this does not establish symmetry

B10 [Proposed]: A Kurtosis That Does Not Converge — Task

- ▣ **Model:** B1 [Solved]; notebook section B10
- ▣ **Data:** r_{sp} ; window n = the returns dated before 3 January of year $1990 + n$, $n = 1, \dots, 36$
 - ▶ 253 returns for $n = 1$, 2 527 for $n = 10$, 9 067 for $n = 36$
- ▣ **Simulation:** 500 Student- t paths of length 9 245 for $\nu = 3$ and for $\nu = 6$ (`default_rng(42)`)
 - ▶ cut at the same window counts
- ▣ **Tasks**
 1. Compute the excess kurtosis of the S&P 500 on each window.
 2. List the three largest jumps, with the days that cause them.
 3. For each ν compute the 5%, 50% and 95% quantiles of the simulated kurtosis at each n .
 4. Plot the data curve over both bands on a symmetric log scale.
- ▣ **Interpretation**
 - ▶ why does the $\nu = 6$ band settle slowly although the kurtosis of t_6 is finite?
 - ▶ what does “excess kurtosis = 10.9” then mean?

B14 [Proposed]: Stylised Facts Reproduced by GARCH- t — Task (1/2)

▣ **Model:** B3 and B11 [Solved], A9; notebook section B14

▣ **Data:** r_{sp} converted to % ($T = 9\,245$)

- ▶ constant mean
- ▶ ω in %² a day

```
from arch import arch_model
from arch.univariate import StudentsT
am = arch_model(100 * r_sp, mean='Constant', vol='GARCH', p=1, q=1, dist='t')
res = am.fit(dis='off')
am.distribution = StudentsT(seed=np.random.default_rng(42)) # reproducible paths
sim = am.simulate(res.params, nobs=len(r_sp), burn=1000)['data']
```

B14 [Proposed]: Stylised Facts Reproduced by GARCH- t — Task (2/2)

□ Tasks

1. Fit the model and list all five parameters with units.
2. Pilot: simulate one path and compute its excess kurtosis.
3. Then simulate 200 paths.
4. For each path compute these statistics.
 - the excess kurtosis
 - the ACF of $|r_t|$ at lags 1, 20, 100, 250
 - the Hill $\hat{\alpha}$ of $|r_t|$ at $k = 2\%$ of T
 - $L(1) = \text{corr}(r_t, |r_{t+1}|)$
5. Build the pointwise 5–95% intervals.
6. Mark each observed value as inside or outside.

□ Interpretation

- ▶ which facts are not reproduced?
- ▶ which model component would you change for each?

B5 [Proposed]: Leverage Effect, BET-TR versus Bitcoin — Task (1/2)

- ▣ **Model:** B3 [Solved]; notebook section B5
- ▣ **Lag convention:** $L(k) = \text{corr}(r_t, |r_{t+k}|)$ Christie (1982)
 - ▶ k counts observation days of each series
 - ▶ $k > 0$: today's return and a later size
 - ▶ example, $k = 1$: the BET-TR return of Monday 16 Mar 2020 is paired with $|r|$ of Tuesday 17 Mar 2020
- ▣ **Data:** `r_bet` and `r_btc`, each on its own calendar
- ▣ **Tasks**
 1. Compute $L(k)$ for $k = -20, \dots, 20$.
 2. Plot both series with the i.i.d. reference bands $\pm 1.96/\sqrt{T}$ (0.036 and 0.030).

B5 [Proposed]: Leverage Effect, BET-TR versus Bitcoin — Task (2/2)

▣ **Tasks** (continued)

3. Report $L(1)$ and $L(5)$ for both markets.
4. State whether they lie outside the reference bands.
5. Give two competing explanations for a weaker Bitcoin effect.

▣ **Interpretation**

- ▶ why is a reference band not a confidence interval under clustering?
- ▶ what does B12 add?

B12 [Proposed]: The Leverage Effect as a Regression — Task (1/2)

- ▣ **Model:** B5, B9 [Solved]; notebook section B12
- ▣ **Regression:** $|r_{t+1}| = c + \beta^+ r_t^+ + \beta^- r_t^- + \sum_{j=1}^5 \delta_j |r_{t-j}| + u_{t+1}$
 - ▶ $r_t^+ = \max(r_t, 0) \geq 0$, $r_t^- = \min(r_t, 0) \leq 0$
 - ▶ a gain $+a$ adds $\beta^+ a$, a loss $-a$ adds $-\beta^- a$
 - ▶ a symmetric response means $\beta^+ = -\beta^-$
 - ▶ design row for day t : $(1, r_t^+, r_t^-, |r_{t-1}|, \dots, |r_{t-5}|)$
 - ▶ response: $|r_{t+1}|$
- ▣ **Data:** r_sp, r_bet, r_btc, r_eur

B12 [Proposed]: The Leverage Effect as a Regression — Task (2/2)

▣ Tasks

1. Estimate by OLS with Newey–West standard errors [Newey & West \(1987\)](#), lag truncation $\lfloor 4(T/100)^{2/9} \rfloor$.
2. Test $H_0: \beta^+ + \beta^- = 0$ at 5%, against χ^2_1 , with the statistic
 - $W = (\hat{\beta}^+ + \hat{\beta}^-)^2 / [\text{Var}(\hat{\beta}^+) + \text{Var}(\hat{\beta}^-) + 2 \text{Cov}(\hat{\beta}^+, \hat{\beta}^-)]$
3. Plot the predicted $|r_{t+1}|$ after a gain $+a$ and a loss $-a$, lag controls at their means.
4. Show why $|r_t|$ cannot be added as a control.

B6 [Proposed]: EUR/RON Across Regimes — Task (1/2)

- ▣ **Model:** B1, B9 [Solved]; notebook section B6
- ▣ **Data:** r_{eur} (BNR reference rate), split into three periods
 - ▶ Jul 2005 – 2011, 2012–2019, 2020–2026: 1 656, 2 013, 1 683 returns
 - ▶ lags built inside each period
- ▣ **Tasks**
 1. In each period regress $r_t = a + \rho r_{t-1} + e_t$ by OLS, with a HAC SE (lag truncation $\lfloor 4(T/100)^{2/9} \rfloor$).
 2. Report $\hat{\rho}$, the SE, the volatility and the excess kurtosis of each period.

B6 [Proposed]: EUR/RON Across Regimes — Task (2/2)

▣ **Tasks** (continued)

3. Test equal ρ across periods with $W = \sum_j w_j (\hat{\rho}_j - \bar{\rho})^2 \sim \chi^2_2$.
 - weights $w_j = 1/\text{SE}_j^2$
 - $\bar{\rho} = \sum_j w_j \hat{\rho}_j / \sum_j w_j$
4. Separately, locate one variance break with the cumulative sum of squares (CUSUM) [Inclán & Tiao \(1994\)](#).
 - $D_k = C_k / C_T - k / T$, with $C_k = \sum_{t \leq k} x_t^2$
 - the break is at $\arg \max_k |D_k|$

▣ **Interpretation**

- ▶ why is the 2020–2026 SE so large?
- ▶ why is the CUSUM date descriptive rather than a test?

B15 [Proposed]: Long Memory or Breaks? — Task (1/2)

- ▣ **Model:** B9, B11 [Solved]; notebook section B15 (helper `local_whittle(x, m)`)
- ▣ **Method Robinson (1995):** demean $x_t = |r_t|$
 - ▶ periodogram $I(\lambda_j)$ at $\lambda_j = 2\pi j/T$
 - ▶ $\hat{d}$ minimises $\ln\left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j)\right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j$ over $d \in [-0.49, 0.99]$
 - ▶ $m = \lfloor T^p \rfloor$; $SE = 1/(2\sqrt{m})$
- ▣ **Data:** $|r_t|$ of the S&P 500, BET-TR and Bitcoin

B15 [Proposed]: Long Memory or Breaks? — Task (2/2)

▣ Tasks

1. Estimate d with $p = 0.65$ and give the 95% interval.
2. Plot $\hat{d}$ against p from 0.45 to 0.80, with the lines $d = 0$ and $d = 0.5$.
3. Locate one variance break of r_t (B6).
4. Re-estimate $\hat{d}$ on each side of the break.
5. Plot the ACF of $|r_t|$ before and after the break.

▣ Interpretation

- ▶ is $\hat{d} > 0$ robust to p and to the break?
- ▶ or is it consistent with regimes [Diebold & Inoue \(2001\)](#)?
- ▶ what does $\hat{d} \geq 0.5$ imply?

B7 [Proposed]: BET-TR vs S&P 500, Same Period — Task (1/2)

- ▣ **Model:** A1, B1 [Solved]; notebook section B7
- ▣ **Data:** BET-TR (total return, RON) and S&P 500 (price index, USD)
 - ▶ align the *prices* on common dates (23 Sep 2014 – 18 Sep 2026)
 - ▶ then compute 2 903 returns
 - ▶ interval lengths: 2 231 one-day, 538 of two or three days, 134 longer (at most 6 days)
- ▣ **Tasks**
 1. Plot the rebased wealth (log scale) and the two underwater curves.

B7 [Proposed]: BET-TR vs S&P 500, Same Period — Task (2/2)

▣ **Tasks** (continued)

2. Compute these measures for both indices.

- the CAGR and the volatility of simple returns ($P = 252$)
- the Sharpe ratio $252\bar{R}/(\sqrt{252}\hat{\sigma}_R)$ with $r_f = 0$
- the Sortino ratio (downside deviation over all days) [Sortino & van der Meer \(1991\)](#)
- the MDD and the Calmar ratio

▣ **Report:** the chart and a two-column table

▣ **Interpretation:** list three reasons why the comparison is descriptive only (dividends, currency, risk-free rate in RON)

B13 [Proposed]: Significance of the Sharpe Difference — Task (1/2)

- ▣ **Model:** B7, B1(d); notebook section B13
- ▣ **Data:** the 2 903 paired returns of B7
 - ▶ simple returns for Sharpe ratios (1.36 and 0.75)
 - ▶ log returns for correlations
- ▣ **Tasks**
 1. Convert to daily Sharpe ratios ($\widehat{SR}/\sqrt{252}$).
 2. Compute the Jobson–Korkie test [Jobson & Korkie \(1981\)](#) with the correction of [Mommel \(2003\)](#).
 - $z = (\widehat{SR}_1 - \widehat{SR}_2)/\sqrt{\widehat{V}}$
 - $\widehat{V} = \frac{1}{T} [2(1 - \hat{\rho}) + \frac{1}{2}(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\hat{\rho}^2)]$

B13 [Proposed]: Significance of the Sharpe Difference — Task (2/2)

▣ **Tasks** (continued)

3. Run a paired circular block bootstrap, 2000 resamples (`default_rng(42)`).
 - draw $\lceil T/21 \rceil$ start days, take 21 consecutive days from each (wrapping at the end), cut to T
 - apply the **same** indices to both series
4. Report the 95% percentile interval of the annualised difference.
5. Compute four correlations.
 - same day
 - BET-TR with the S&P 500 of the previous day and of the next day
 - weekly (Friday) returns

▣ **Interpretation**

- ▶ is the difference significant?
- ▶ what do both methods assume?
- ▶ which correlation belongs in a portfolio?

B8 [Proposed]: Range-Based Estimators on the S&P 500 — Task (1/2)

- ▣ **Model:** A5 [Solved]; notebook section B8
 - ▶ tested helpers for all five estimators
 - ▶ formulas in [Chapter 1](#)
- ▣ **Data:** S&P 500 index OHLC, 2 Jan 2008 – 18 Sep 2026
- ▣ SPY ETF on one adjusted basis
 - ▶ C^* = the adjusted close (for dividends and splits)
 - ▶ $O^*, H^*, L^* = (C^*/C) \times O, H, L$
 - ▶ example, 16 Mar 2020: factor $218.08/239.85 = 0.9092$; $O^* = 241.18 \times 0.9092 = 219.29$, $H^* = 233.58$, $L^* = 215.82$

B8 [Proposed]: Range-Based Estimators on the S&P 500 — Task (2/2)

▣ Tasks

1. Compute 21-day rolling close-to-close, Parkinson, Garman–Klass [Garman & Klass \(1980\)](#), Rogers–Satchell [Rogers & Satchell \(1991\)](#) and Yang–Zhang [Yang & Zhang \(2000\)](#) volatilities ($P = 252$).
2. Compute their time averages.
3. For the index and for SPY, compare the 21-day averages of close-to-close and Yang–Zhang with two full-sample volatilities.
 - overnight: $\text{sd}(o_t - c_{t-1})\sqrt{252}$
 - open-to-close

- ▣ **Interpretation:** why does Yang–Zhang stay below close-to-close on the index but not on SPY?

C1 [Proposed]: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? (1/2)

- ▣ **Model:** B6, B9 [Solved]; a project seed
- ▣ **Context:** BET-TR has a positive lag-1 autocorrelation, typical of less liquid markets
 - ▶ Romania became a FTSE Russell Secondary Emerging market on 21 Sep 2020
- ▣ **Data:** r_{bet} and a liquidity measure
 - ▶ liquidity: daily traded value (volume $\times$ raw close, in lei) of 11 BVB blue chips
 - ▶ the stocks: TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL
 - ▶ summed over the stocks listed that day
 - ▶ report how many are listed each year

C1 [Proposed]: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? (2/2)

▣ Tasks

1. Compute $\hat{\rho}_1$ on rolling 250-day windows (step 5 days).
2. Add the B9 bands, labelled as reference bands under $\rho_1 = 0$.
3. Estimate ρ_1 before and after 21 Sep 2020 as an AR(1) slope with HAC SEs (B6).
4. Report $z = (\hat{\rho}_{\text{after}} - \hat{\rho}_{\text{before}}) / \sqrt{SE_1^2 + SE_2^2}$.
5. Report the interaction e in $r_t = a + b r_{t-1} + c \text{post}_t + e r_{t-1} \text{post}_t$ (HAC, 10 lags).
6. Correlate the rolling $\hat{\rho}_1$ with the log 250-day average traded value.

▣ Deliverable: a Quantlet folder and one slide

- ▶ a descriptive conclusion and its main limitation
- ▶ not a causal claim

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (1/4)

- **Model:** the lecture case study
 - ▶ a replication project, not an in-class task
- **Question:** can past S&P 500 returns alone explain the VIX, as [Guyon & Lekeufack \(2023\)](#) find?
 - ▶ 94.6% of its variance in 2000–2018, 85.5% in 2019 – May 2022
- **Data:** daily closes of the S&P 500, the VIX and the 9-day VIX
 - ▶ simple returns $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$
 - ▶ volatility indices divided by 100
- **Model, eq. (3.7) in the paper:** $\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t$
 - ▶ $R_{1,t} = \sum_{i=0}^{999} K_1(i\Delta t) r_{t-i}$
 - ▶ $\Sigma_t = \left(\sum_{i=0}^{999} K_2(i\Delta t) r_{t-i}^2 \right)^{1/2}$
 - ▶ $\Delta t = 1/252$

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (2/4)

▣ **Kernels**, eq. (3.9) in the paper: $K_j(\tau) = (\tau + \delta_j)^{-\alpha_j} / Z_j$

▶ Z_j such that $\sum_{i=0}^{999} K_j(i\Delta t)\Delta t = 1$

▣ **Parameters and units**

▶ β_0 in volatility units ($0.03 = 3$ VIX points)

▶ β_1, β_2 unitless

▶ $\alpha_j > 1$, $\delta_j > 0$ in years

▣ **Start** (Table 3 in the paper, VIX row)

▶ $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$

▶ $\alpha_1 = 1.06$, $\delta_1 = 0.020$, $\alpha_2 = 1.60$, $\delta_2 = 0.052$

▶ bounds $\alpha_j \in [1.0001, 20]$, $\delta_j \in [10^{-4}, 5]$

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (3/4)

▣ **Tasks**, in stages

1. Fixed kernels: compute $R_{1,t}$ and Σ_t with the Table 3 kernels.
2. Estimate $\beta_0, \beta_1, \beta_2$ by OLS on 2000–2018.
3. Full model: estimate all seven parameters by nonlinear least squares on 1 Jan 2000 – 31 Dec 2018.
 - use `scipy.optimize.least_squares`
 - recompute the features at every trial value
4. Report r^2 and RMSE on the training set and on 1 Jan 2019 – 15 May 2022.
 - for the full model and for the restrictions $\beta_1 = 0$ and $\beta_2 = 0$
5. Repeat the full model for the 9-day VIX (training from 2011).
6. Keep the parameters fixed and score 16 May 2022 – 18 Sep 2026.

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (4/4)

▣ **Report**

- ▶ the parameter table and the scores next to Table 2 of the paper
- ▶ the observed and fitted VIX, and the ratio of observed to fitted VIX
- ▶ the fitted kernels

▣ **Interpretation:** the model fits the VIX on the same day

- ▶ it is not a next-day forecast
- ▶ which feature explains more, $R_{1,t}$ or Σ_t ?

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